

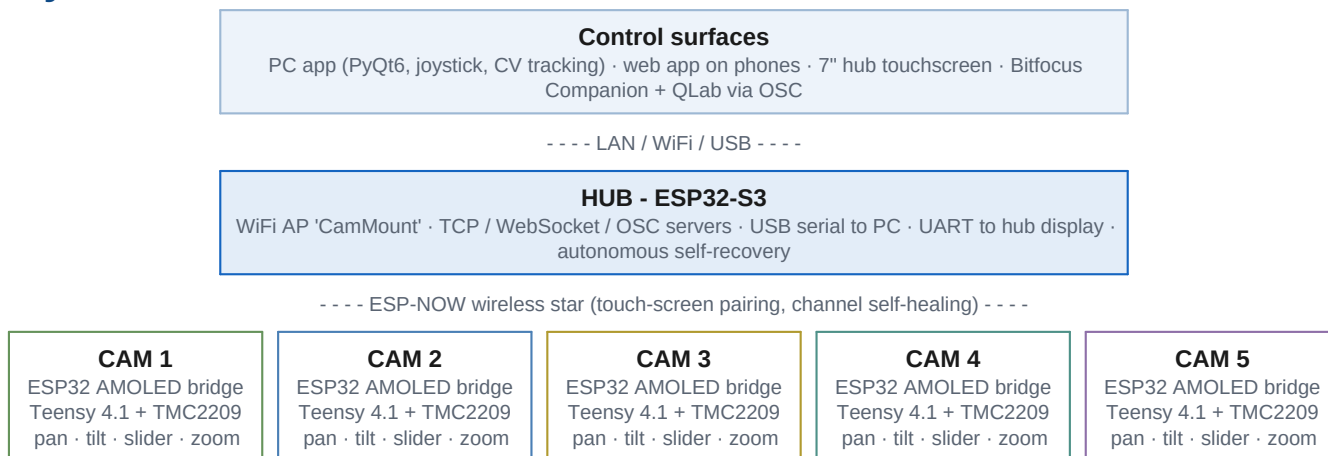
PTS

Pan / Tilt / Slider

A five-mount motorised camera system for live video production - wireless, self-healing, and controllable from everything in the booth.

PTS drives five motorised camera mounts over a dedicated wireless network. Each mount pairs a Teensy 4.1 motion controller (TMC2209 silent stepper drivers, StallGuard homing) with an ESP32 bridge and round touchscreen; a central hub links every mount to the operator's control surfaces. Positions, speed presets and subject calibrations live on the mounts themselves - controllers can come and go mid-show.

System architecture



Highlights

Feature	What it does
Look-at tracking	Camera keeps aiming at a calibrated 3D subject while the slider travels - subjects are calibrated in two taps and switchable mid-move from any controller.
Zero-config pairing	No MAC addresses or per-unit firmware: hold a mount's screen, pick CAM 1-5, tap your hub. The hub learns mounts on first contact; conflicts resolve with one tap on the hub display.
Self-healing	Autonomous recovery ladders on hub and mounts (reinit then restart), idle maintenance restarts, watchdogs at every layer, and a motion-side dead-man: a mount stops within 500 ms of losing its control stream.
Health telemetry	Every node reports heap, loop timing, link quality and reset cause every 10 s into the PC log - degradation is visible hours before it becomes a symptom.
Show-control ready	OSC servers on both the hub and the PC app put every function on Stream Deck buttons and QLab cue stacks (reference overleaf).

OSC control reference - Companion / QLab

Two OSC servers listen on **UDP port 9700** and accept the same commands, so Companion pages fire unchanged against either target. **Feedback is sent by the hub only** - point Companion at the hub for buttons that light up to show what the rig is doing:

Target	Address	When to use
The hub itself	192.168.4.1	No PC needed - hub + display + mounts + Stream Deck is a complete rig. Reach it via a WiFi-to-LAN bridge joined to the CamMount AP, or any machine joined to CamMount directly.
PC app machine	that PC's IP	When the PC app is running anyway. Configurable via osc_enabled / osc_port in the app config.

Companion: add a **Generic: OSC** connection with the target IP and port 9700, then use its Send message actions as below. Integer arguments preferred; floats (QLab network cues) are accepted.

Command reference

Address	Arguments	Action
/pts/estop	-	E-STOP every mount
/pts/cam/N/estop	-	E-STOP mount N
/pts/cam/N/goto	slot 1-10	Recall a stored position (uses the mount's active speed presets)
/pts/cam/N/store	slot 1-10	Store the current position
/pts/cam/N/clear	slot 1-10	Clear a stored position
/pts/cam/N/jog	pan tilt slider zoom	Velocities -1000..1000; re-streamed at 20 Hz until stopped
/pts/cam/N/jog/stop	-	Stop jogging (put on the button's release action)
/pts/cam/N/jog/left /right /up /down	1 press, 0 release	Pan and tilt from plain buttons; full deflection at the active preset speed
/pts/cam/N/jog/slide/l eft /slide/right	1 / 0	Slide along the rail
/pts/cam/N/jog/zoom/in /zoom/out	1 / 0	Zoom
/pts/cam/N/speed/pt	1-4	Active pan/tilt speed preset
/pts/cam/N/speed/sl	1-4	Active slider speed preset
/pts/cam/N/speed/pt/up /down	-	Pan/tilt preset one step, clamped at 1 and 4
/pts/cam/N/speed/sl/up /down	-	Slider preset one step, clamped
/pts/cam/N/speed/pt/in c /sl/inc	-	Cycle the preset 1-2-3-4-1
/pts/cam/N/subject	0-7	Select look-at subject; switches live during a move
/pts/cam/N/lookat	0 or 1	Look-at slider move to min (0) or max (1) with the selected subject
/pts/refresh	-	Resend all feedback for every mount
/pts/cam/N/refresh	-	Resend all feedback for mount N

N = camera 1-5. Slots and speed presets are 1-based, subjects 0-based - matching every screen in the system.

Feedback - buttons that show the rig

The hub sends state back to whoever last commanded it, so a Stream Deck shows which shot a camera is on, whether a mount is online and what speed it is set to - with nothing tracked in Companion. Enable **Listen for Feedback** on the OSC connection, then add an *OSC: Listen for OSC messages (Integer)* feedback matching an address and value below.

Address	Value	Meaning
<code>/pts/cam/N/active</code>	0 / 1	Mount is online
<code>/pts/cam/N/state</code>	int	Mount state; 0 is idle
<code>/pts/cam/N/target</code>	0-10	Slot being moved to, 0 = not moving to one
<code>/pts/cam/N/speed/pt</code>	1-4	Active pan/tilt preset
<code>/pts/cam/N/speed/sl</code>	0-4	Active slider preset; 0 = this mount has no rail
<code>/pts/cam/N/slot/M/state</code>	0-3	0 empty, 1 occupied, 2 moving here, 3 arrived

Slot state is one address per slot carrying one value, so a shot button colours itself from a single feedback with nothing to combine - the same stored / moving-to / arrived picture the hub display and the web app show. Only changes are sent, so a rig at rest is silent; everything is resent every 5 s, and in full the first time an address speaks to the hub, so a surface that joins late catches up on its own. **If nothing arrives**, check Companion's Source Port is the port it also listens on, and that its machine is on the same subnet as the hub - commands still arrive across a subnet mismatch, replies cannot get back, and UDP reports no error either way.

Button recipes

Button	Setup
Recall shot 3, cam 2	Press: <code>/pts/cam/2/goto</code> with int 3
Hold-to-jog pan-left, cam 1	Press: <code>/pts/cam/1/jog/left</code> int 1 Release: same address, int 0
Shot button that shows its state	Press: <code>/pts/cam/1/goto</code> int 7. Feedback on <code>/pts/cam/1/slot/7/state</code> : 1 red (stored), 2 amber (on its way), 3 green (arrived).
One speed button per camera	<code>/pts/cam/1/speed/pt/inc</code> cycles 1-2-3-4-1 per press. Add a feedback on <code>/pts/cam/1/speed/pt</code> to show which preset is live.
Look-at pair, cam 3	Button A: <code>/pts/cam/3/subject</code> int 2 Button B: <code>/pts/cam/3/lookat</code> int 1 (press A mid-move to retarget live)
QLab cue	Network cue to port 9700: <code>/pts/cam/2/goto</code> 4 - camera moves fire from the cue stack, in time with light and sound.
Show-stopper	A big red <code>/pts/estop</code> (no argument).

Safety

- OSC jogs are re-streamed at 20 Hz with a 15 s TTL, so a lost release message cannot run a mount away; prefer `goto` / `lookat` for long moves.
- The mounts' own 500 ms dead-man is the final backstop: if the hub or PC dies mid-jog, motion stops regardless of the controller.
- E-STOP over OSC clears held jog streams before broadcasting the stop.